

THE PHOSPHATE DEPOSITS OF SOUTH CAROLINA.

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INTRODUCTION.

The deposits of phosphate rock in the vicinity of Charleston, S. C., were among the first discovered in the United States, and until about 20 years ago the field was one of the largest producers in the world. The workable deposits are confined to the region lying in general between Charleston and Beaufort and extending about 25 miles back from the coast. The phosphate rock occurs in a thin bed which underlies irregular areas within the region defined. This bed is commonly less than 18 feet below the surface, and is thus comparatively easy to mine, but, owing to the rather low grade of the rock and the exploitation of richer and more easily accessible deposits in Florida and Tennessee, production in the South Carolina field has now almost ceased. A number of papers describing this field have been published, most of them 20 years or more ago, but in view of the present more extended geologic information concerning this region it has been thought advisable to prepare this account, which is based on a brief field study by the writer.

The examination of the field was made in January, 1914. The writer is indebted to Mr. Earle Sloan, formerly State geologist of South Carolina, for many courtesies and for some unpublished information concerning the geology. Mr. John Hertz, who has long been engaged in phosphate mining, kindly contributed several maps and furnished valuable data, and Messrs. William Gaillard, Bachman Smith, and William Alston assisted in many ways. In addition to information privately communicated, several of the earlier reports on the field have been freely drawn upon for pertinent facts.

The earliest report on the phosphate deposits was published in 1868 by N. A. Pratt,¹ and this was followed in 1870 by a more extended account by F. S. Holmes.² Both of these papers are written in narrative and more or less popular form. In 1870 N. S. Shaler³ contributed a short scientific description. In 1880 C. U. Shepard, jr.,⁴

¹ Ashley River phosphates, Philadelphia, 1868.

² Phosphate rocks of South Carolina, Charleston, 1870.

³ On the phosphate beds of South Carolina: Boston Soc. Nat. Hist. Proc., vol. 13, pp. 222-235, 1870: U. S. Coast Survey Rept., pp. 182-189, 1870.

⁴ South Carolina phosphates, Charleston, 1880.

published the most complete and valuable of the earlier reports on the field. In 1888 R. A. F. Penrose, jr.,¹ wrote for the United States Geological Survey a bulletin on phosphate which contains a good though brief account of the South Carolina deposits. In 1891 a book on American phosphate by Francis Wyatt² appeared, in 1892 a book on phosphate by C. C. H. Millar,³ and in 1895 a study of the phosphate industry by David Levat,⁴ all three of which deal with the South Carolina field at greater or less length, but discuss chiefly the mining and product costs. Charles L. Reese⁵ published a short paper on the region of the deposits based on a series of chemical experiments. An extended account of the field and narrative of its discovery and history, prepared primarily for newspaper publication, was written by P. E. Chazal⁶ in 1904. In 1908 Earle Sloan,⁷ State geologist, published a "Catalogue of the mineral localities of South Carolina," which contains geologic sections measured at a number of places and a short account of the phosphate. In 1913 William H. Waggaman,⁸ of the United States Department of Agriculture, wrote a brief report on the field, dealing chiefly with the state of the industry and methods of production. In addition to these general papers, an elaborate description by Joseph Leidy⁹ of the vertebrate fossils found in the phosphate beds appeared in 1881, and a brief account of some of the invertebrate forms by W. H. Dall¹⁰ in 1894.

GEOGRAPHY AND TOPOGRAPHY.

The phosphate deposits are confined to a seaboard zone, most of it less than 30 miles in width, which extends from Beaufort on the south to a point about 20 miles beyond Charleston on the north. (See Pl. II.) The most important deposits, however, are situated between the mouth of Broad River and a point near the source of Wando River and lie within a rough arc terminating at these points and extending farthest back along Edisto River. Within this segment the phosphate underlies irregular areas whose boundaries have not been exactly defined but which are approximately delimited on the accompanying map.

The principal cities within this area are Charleston and Beaufort, the former being the center of the phosphate and fertilizer industry.

¹ Nature and origin of deposits of phosphate of lime: U. S. Geol. Survey Bull. 46, pp. 60-70, 1888.

² Phosphates of America, pp. 44-61, New York, 1891.

³ Florida, South Carolina, and Canadian phosphates, pp. 123-178, London, 1892.

⁴ Industrie des phosphates et superphosphates, pp. 83-90, Paris, 1895.

⁵ On the influence of swamp waters in the formation of the phosphate nodules of South Carolina: Am. Jour. Sci., 3d ser., vol. 43, pp. 402-406, 1892.

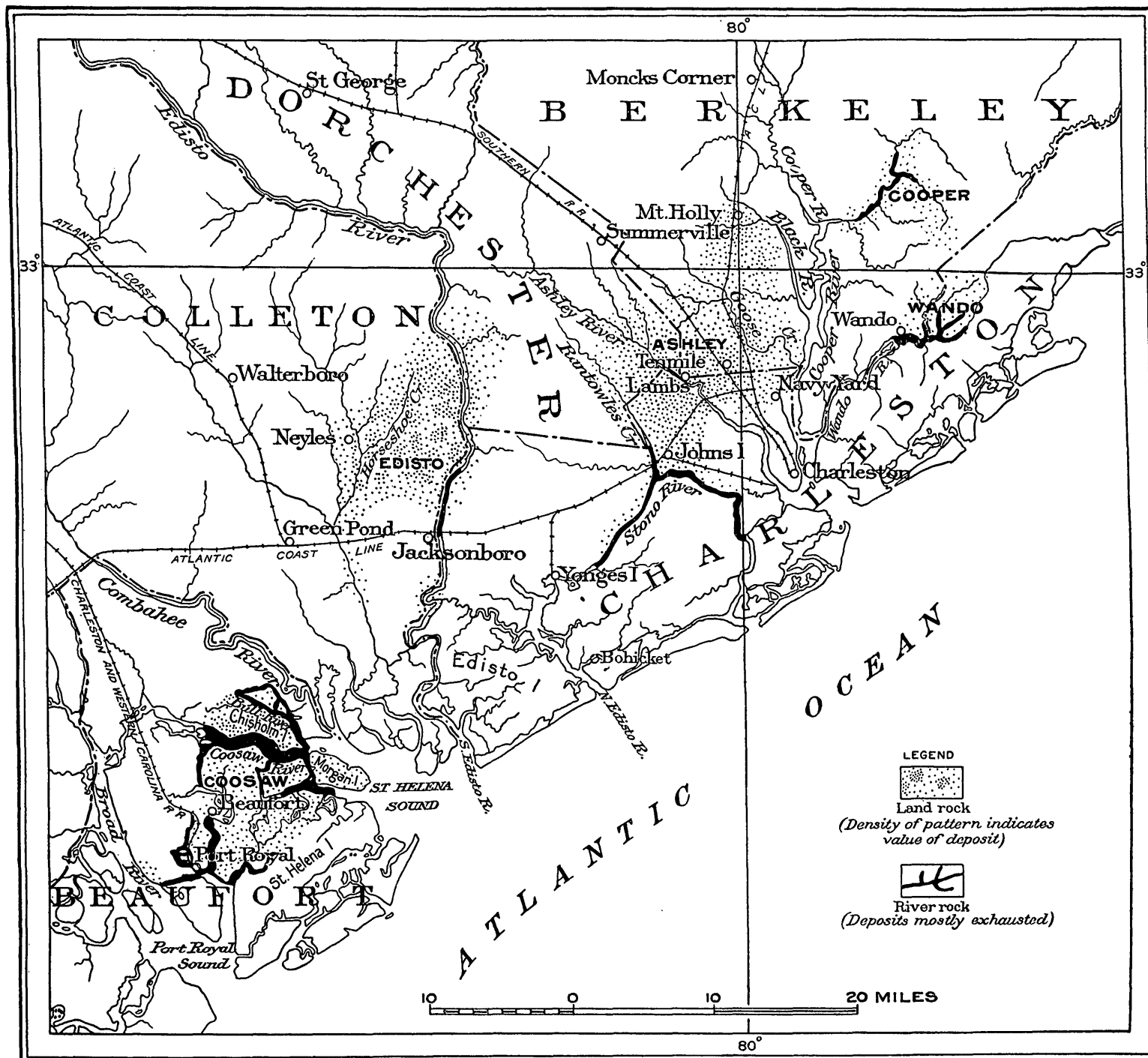
⁶ Sketch of the South Carolina phosphate industry, Charleston, 1904.

⁷ South Carolina Geol. Survey, 4th ser., Bull. 2, 1908.

⁸ Report on the phosphate fields of South Carolina: U. S. Dept. Agr. Bull. 18, 1913.

⁹ Description of vertebrate remains, chiefly from the phosphate beds of South Carolina: Jour. Acad. Nat. Sci. Philadelphia, vol. 8, 2d ser., pp. 209-262, 1874-1881.

¹⁰ Notes on the "land phosphate" of the Ashley River district, S. C.: Am. Jour. Sci., 3d ser., vol. 48, pp. 300-301, 1894.



MAP SHOWING APPROXIMATE ORIGINAL DISTRIBUTION OF SOUTH CAROLINA PHOSPHATE DEPOSITS.

In previous years a large amount of phosphate rock was shipped from these points or from docks located farther up the rivers in their vicinity. The Atlantic Coast Line, the Southern Railway, and the Charleston & Western Carolina Railway cross the area, and most of the phosphate properties are located either on the main lines or on short spurs of these roads.

The coast in this vicinity is low and extremely irregular in outline and is bordered by numerous islands. It is intersected in the most intricate manner by numberless tidewater channels which range in width from a few feet up to several miles, one of the widest being Broad River at Port Royal. Many of these channels run almost parallel to the coast and thus form islands, and in some localities they have intersected the coast to such an extent that two or three tiers of islands have been formed.¹ In addition to these arms of the sea, many of which extend considerable distances back from the coast proper, some of the streams and rivers that traverse this region have developed a system of tidewater distributaries. Edisto, Ashepoo, and Broad rivers are conspicuous examples of streams of this type, which, in conjunction with the waterways mentioned above, have intricately dissected the general coastal region. (See Pl. II.) Most of the rivers are broad, sluggish streams and are navigable for considerable distances above their mouths, affording ready means of transportation for the phosphate.

The topography within the whole phosphate-bearing area is very uniform, being everywhere of the Coastal Plain type. The land is almost flat and is for the most part less than 15 feet above tide; elevations of more than 25 feet are uncommon. Most of it is wooded and is swampy in character, being practically covered with water for at least part of the year. In the course of the phosphate mining during the last 45 years many large areas have been worked over with steam shovel or by hand, and these tracts, in which the surface is broken, swampy, and covered with dense thickets, are now practically impassable.

GEOLOGY.

STRATIGRAPHY.

SEQUENCE OF THE STRATA.

Owing to the wealth of fossil remains which some of the strata in this region contain they have attracted the attention of many observers. Ruffin,² Tuomey,³ Pratt,⁴ Holmes,⁵ and Shepard,⁶ all give more

¹ A good account of the configuration of the coast, the submarine topography, etc., is given by N. S. Shaler in U. S. Coast Survey Rept., 1870, pp. 182-185.

² Ruffin, Edmund, Agricultural survey of South Carolina, Columbia, 1843.

³ Tuomey, Michael, Report on the geology of South Carolina, Columbia, 1848.

⁴ Pratt, N. A., Ashley River phosphates, Philadelphia, 1868.

⁵ Holmes, F. S., Phosphate rocks of South Carolina, Charleston, 1870.

⁶ Shepard, C. U., jr., South Carolina phosphates, Charleston, 1880.

or less extended descriptions of the geology. Their work has been largely superseded, however, by the detailed investigations of Sloan¹ and the broad areal studies of Vaughan.² In conformity with the results of this recent work the following classification has been adopted by the United States Geological Survey for the formations treated in this report:

Geologic formations related to the phosphate deposits of the Charleston region, S. C.

Sys-tem.	Series.	Formation.	Approximate thickness.	Character.
Quar-ter-nary. Ter-ti-ary.	Pleistocene.	Undifferentiated.	<i>Feet.</i> Variable.	Sand, gravel, loam, etc.
	Pliocene.	Unconformity —		
		Waccamaw marl.	11	Compact granular marl; upper part brownish yellow, lower part gray.
	Miocene.	Unconformity —		
		Duplin marl.	35	Yellow marl; upper part very hard, characterized by <i>Pecten eboreus</i> ; middle compact to porous, containing <i>Chama striata</i> and <i>Arca incile</i> ; lower friable and porous, containing <i>Amusium mortoni</i> .
		Unconformity —		
		Edisto marl.	0-4	Marl, commonly more or less phosphatized; compact in texture, but containing many large cavities; color light where unaltered, darker where phosphatized; source of the commercial phosphate.
	Oligocene.	Unconformity — Apparently unrepresented in this area.		
	Eocene.	Cooper marl.	100+	Marl, light grayish green to dark greenish drab.

TERTIARY SYSTEM.

EOCENE SERIES.

Cooper marl.—The Cooper marl is the lowest formation here considered, though earlier Eocene is well developed within the general phosphate-bearing area. The Cooper conformably overlies the Barnwell sand (equivalent in part to the Mount Hope marl of Sloan), which is the highest formation of the Claiborne group in this region. The Cooper is therefore in whole or in part referable to the Jackson group, the highest division of the Eocene, but diagnostic fossils are rare and the exact age of the upper part of the formation has not yet been determined. The Cooper is, however, almost entirely late Eocene, but its higher strata may have been deposited in early Oligocene time.

The formation consists of over a hundred feet of grayish-green marl; the lower part (the Cooper marl of Sloan) is greenish drab in color and somewhat plastic when wet, but lighter colored and fairly

¹ Sloan, Earle, South Carolina Geol. Survey, 4th ser., Bull. 2, 1908.

² For discussion by Vaughan see Willis, Bailey, Index to the stratigraphy of North America: U. S. Geol. Survey Prof. Paper 71, pp. 806-813, 1912.

hard when dry. It commonly contains about 75 per cent lime carbonate and about 2 per cent lime phosphate. The upper part of the formation (the Ashley marl of Sloan) is dull olive green and semiplastic when wet and drab when dry. The content of lime carbonate is somewhat lower than that of the lower division of the formation, but the phosphate is considerably higher, commonly exceeding 5 per cent. (See analyses, p. 208.)

Shepard¹ gives an interesting well section made at Sineaths station, which presumably includes the greater part of the Cooper and probably also part of the underlying Claiborne group. The section is, unfortunately, not complete, but gives only the depth of certain strata which Shepard analyzed for phosphate. (See analyses, p. 208.) It shows, however, that thin layers of large nodules containing 50 per cent or more of phosphate occur at depths of 26, 70, and 110 feet below the surface, and that the inclosing marl contains phosphate in amounts varying from a trace to 3 per cent.

The exact thickness of the Cooper marl has not been determined. The Cooper is being quarried at Woodstock, about 15 miles north of Charleston. At the time of the writer's visit only about 65 feet was exposed, but it is reported that the base of the marl had been reached by drilling to a depth of 140 feet. It is possible, however, that this may include part of an underlying formation. An excellent section of the Cooper is exposed at the Woodstock pit, from which about 5,000 tons a year is quarried for use as a base or "filler" for the commercial fertilizer. The marl is sufficiently coherent to form the vertical walls of the deep pit, yet soft enough to be worked with pick and shovel. It is blasted out, broken up, and transported by bucket hoist to the shed, where it is burned and ground. The marl is compact and impervious to water, so that the pit is fairly dry. Invertebrate fossils are rather scarce, but fragments of bone are not uncommon.

OLIGOCENE SERIES.

In the phosphate area the Oligocene is apparently not represented, and the Edisto marl (Miocene), the source of the phosphate with which this paper is primarily concerned, rests directly on the Cooper marl (Eocene).

It may be pointed out, however, that the absence of Oligocene strata does not necessarily imply that they were never deposited. The Cooper marl may overlap slightly into Oligocene time (see p. 186), and this suggests that there may have been no marked break at the

¹ Shepard, C. U., jr., *Rural Carolinian*, August, 1873. See also Chazal, P. E., loc. cit., and Waggaman, W. H., loc. cit.

end of the Eocene and that deposition may have continued for some time. If it did, then all the strata down to the present top of the Cooper must have been eroded during the subsequent elevation of the land.

MIocene SERIES.

Edisto marl.—In this region the Edisto marl, the source of the phosphate rock, rests directly and unconformably on the Cooper marl, the hiatus representing all or nearly all of the Oligocene as well as the lowest Miocene. Similarly, the Edisto is commonly overlain directly by the Quaternary, although two higher Tertiary formations are well recognized in this general area. Although its identifiable fossils are few the Edisto is believed by Vaughan¹ to be middle Miocene—about equivalent to the St. Marys formation of Maryland and Virginia. The opinions of the older workers as to the age of the Edisto differed widely. Tuomey² referred it to the Eocene; Holmes, Shepard, and Chazal considered it an altered conglomerate of Eocene marl deposited in post-Pliocene time; and Pratt believed it to be Recent. Dall³ in 1894 first definitely established its age as Miocene.

According to Sloan⁴ the Edisto in its unaltered form is a yellow-white marl, commonly 4 to 5 feet thick. Although compact in texture it is penetrated in many places by tortuous cavities, some organic in origin and others apparently due to solution. It has been found in its unaltered form at only a few localities, being commonly phosphatized. The two forms may be observed in close proximity in the bed of Stono River near the mouth of Wappoo Cut. In the phosphatized form the bed is generally thinner, averaging about 1 foot and at few places exceeding 2½ feet. The phosphatized mass is fairly hard and is irregular in outline and honeycombed with many cavities. These may reach 4 inches or more in diameter and in some localities are so large that the rock is mostly broken up into irregular nodules. The filling of the cavities (or of the matrix when the bed is separated into fragments) is a black sandy clay, generally calcareous and phosphatic and in many places carrying abundant vertebrate remains. These are chiefly cetacean bones and sharks' teeth and vertebræ, but the coprolites and bones of late Miocene, Pliocene, and Quaternary land animals are also plentiful. A few distinctively Eocene shells, generally waterworn, are also found in this material. The phosphate rock itself, however, contains numerous casts and molds of the same shells that characterize the unaltered Edisto marl.⁵

In addition to this regularly stratified form the phosphate rock is also found in irregular deposits in the beds of many of the rivers.

¹ U. S. Geol. Survey Prof. Paper 71, p. 807, 1912.

² Tuomey, Michael, op. cit., pp. 164-165.

³ Am. Jour. Sci., 3d ser., vol. 48, pp. 300-301, 1898.

⁴ Sloan, Earle, op. cit., p. 470.

⁵ Idem, p. 299.

This river rock generally consists of rounded pebbles, and is commonly darker in color and higher in specific gravity than the ordinary land rock. It is probable that these deposits were in part laid down during Edisto time, but that they were subsequently worked over by river action and were increased in thickness by the erosion and concentration of the land deposits.

Duplin marl.—The Duplin marl, an extension of the marl of the same name in North Carolina, includes the “Peedee” and “Goose Creek” marls of Sloan. It is upper Miocene in age. The lower part of the Duplin marl (the “Goose Creek marl” of Sloan) is a soft yellow porous marl containing over 80 per cent of carbonate of lime. According to Sloan it overlies the unphosphatized Edisto¹ at Givhams Ferry, and its outcrop near Goodrich station sharply delimits on the south that of the phosphatized Edisto.² At the latter place it contains near the contact nodules of phosphate rock that prove it to be younger, but although it presumably overlies the phosphate to the south it was not actually observed in this position. Except for these two localities the Edisto and Duplin marls have not been observed in the same section and in the main appear to occupy different areas.

The soft yellow porous marl is overlain in places by 25 feet of harder and more compact yellow marl, to which Sloan has applied the name “Peedee marl.” There appears to be no significant faunal or lithologic differences between the two, however, and they are therefore grouped by Vaughan under the name Duplin marl. The upper member (“Peedee marl” of Sloan) has not been observed in contact with the Edisto.

PLIOCENE SERIES.

Waccamaw marl.—The Pliocene series in South Carolina is represented by the Waccamaw marl, which occurs only in Horry County near the North Carolina-South Carolina State line. It is not found in contact with the Edisto, the phosphate-bearing formation, and is here included for the sake of completeness. It is described by Sloan as a compact granular marl about 11 feet thick, brownish yellow at the top and gray beneath.

QUATERNARY SYSTEM.

PLEISTOCENE SERIES.

The Pleistocene deposits have been differentiated by Sloan into a number of subdivisions, but as their broader relations are not well established they are here treated as a unit. Because of the conditions described above, the Pleistocene generally rests directly on the Edisto within the phosphate-bearing area and constitutes the overburden

¹ Sloan, Earle, op. cit., p. 283.

² Idem, p. 291.

which must be removed in mining. It consists of clay, marl, sand, and gravel, 20 feet or more in thickness. This material has been divided by Sloan ¹ as follows:

5. Sea Island loams. Fine, glauconitic sand and silt, probably derived from the reworking of the Bohicket marl sands.

4. Wando clays and sands. White sand overlain by drab to red clay.

3. Accabee gravels, generally absent. Pebbles of quartz and phosphate rock derived from the reworking of the Edisto marl.

2. Bohicket marl sands. Fine-grained, greenish to bluish-green glauconitic sand.

1. Wadmalaw marl. Greenish-gray sandy marl inclosing large numbers of Pleistocene shells.

This classification appears to hold within the phosphate-bearing area, although the complete section is probably not exposed at any one place. In addition Sloan differentiates what he calls the Ten-mile sands, which were deposited in early Pleistocene time in the form of a barrier ridge, on the seaward side of which the above-mentioned deposits were laid down.

Sloan's Wadmalaw marl is in general only 2 to 3 feet thick and is fairly persistent. It is overlapped by his Bohicket marl sands, which extend several miles beyond it to the north and are almost invariably present in the material overlying the phosphate and in places are 10 feet or more thick. These deposits are somewhat glauconitic and at the Bolton mines contain 2.19 per cent of potash. The Accabee gravels of Sloan form an interesting example of the reworking of an older bed. According to Sloan the material was derived from the partial disruption of the shoreward edge of the Edisto marl by wave action, especially in the Wando and Cooper basins. The area of the gravel is very small; where present it is generally less than a foot thick, though in places it attains a thickness of 4 feet. It has afforded a limited supply of phosphate and is known to the miners as the "Flying Rock bed." The Wando clays and sands of Sloan overlie the gravel in places, the sands being widely distributed. Sloan's Sea Island loams are best developed along the zone of the sands to which he applied the name "Bohicket" and are probably derived from them.

The character of the Pleistocene deposits that overlie the Edisto is illustrated by the following sections:

Section at Bolton mine, Charleston County.²

	Feet.
Vegetable muck.....	4
Clay, green, sandy, glauconitic (Bohicket of Sloan).....	5
Sand (Bohicket of Sloan).....	3
Marl, highly fossiliferous (Wadmalaw of Sloan).....	2
Phosphate rock (Edisto).....	2

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¹ Sloan, Earle, op. cit., pp. 480-484.

² Idem, p. 298.

Section near Lambs, Charleston County.

[Measured by G. S. Rogers.]

	Feet.
Vegetable muck.....	1
Sand, white, brownish at top (Wando (?) of Sloan).....	6
Sand, yellow.....	1
Sand, bluish-green, clayey (Bohicket of Sloan).....	3
Phosphate rock (Edisto).....	1

12

Section at Simmons Bluff, Youngs Island, Colleton County.¹

	Ft.	in.
Soil, sandy.....	2	0
Wando (?) of Sloan:		
Loam, red, becoming clayey toward base.....	6	0
Clay, white, sandy, plastic.....	1	6
Sand, white, fine.....	1	6
Bohicket of Sloan:		
Clay, green, glauconitic.....	3	0
Wadmalaw of Sloan:		
Loose shells.....	3+	

17+

STRUCTURE.

The strata in the South Carolina phosphate area have undergone little or no tilting and for the most part lie almost flat. Such variations from the horizontal as they exhibit are due chiefly to the character of the old surface on which they were deposited: All the formations have a slight but gentle dip toward the sea owing to the shelving character of the old surface; and as this surface was in places irregular most of the formations vary considerably in thickness and in some localities are absent.

GEOLOGIC HISTORY.

During late Eocene time the region here considered was covered by the sea and the submergence may have extended into early Oligocene time. During this period the Cooper marl was laid down to a thickness of over 125 feet. At some time during the Oligocene the land was elevated and remained above the sea for a long time, during which a portion of the upper part of the Cooper was removed by erosion. According to Sloan the shore line during this period was roughly coincident with the present position of Wando and Stono rivers. (See Pl. II.) At the opening of the middle Miocene the land was again partly submerged, and the Edisto was deposited on the irregular upper surface of the Cooper. The hills of the old surface were perhaps in part submerged and in part formed low islands, on which the Edisto was never laid down. This formation thus appears to be entirely

¹ Sloan, Earle, *op. cit.*, p. 300.

lacking in the area between Ashepoo and Combahee rivers and in several smaller districts. The old surface was somewhat shelving, as is indicated by the present slight dip (3.3 feet to the mile) of the Edisto toward the sea. Farther seaward this inclination of course increased, and the Edisto therefore dips sharply below the present surface along the southern boundary of the phosphate area, as indicated roughly on the map (Pl. II). Thus, according to Shepard,¹ the Edisto at Charleston is about 60 feet below the surface. After its deposition the Edisto marl underwent certain chemical changes, which increased its phosphate content. (See p. 206.)

At the end of Edisto time a slight elevation of the land took place and the Duplin marl was laid down within the area which still remained below water. A further elevation of land occurred at the end of Miocene time, and no more marine sediment was laid down in the immediate phosphate area until the Pleistocene.

According to Sloan the earliest Pleistocene formation is his Ten-mile sands, which were deposited in the form of a barrier ridge along a zone near the north border of the phosphate area. During the gradual subsidence which followed, the deposits called by Sloan the Wadmalaw, Bohicket, and Accabee were deposited against the seaward side of this ridge as a progressively overlapping series. The relations of the deposits to which he gave the names Wando and Sea Island are less clear, but they appear to have been laid down during the gradual reelevation which doubtless brought the land to its present altitude.

PHOSPHATE ROCK.

PHYSICAL CHARACTER.

The phosphate was described by Pratt in 1868 and by Shepard in 1880 as essentially nodular in character, and this description has apparently been followed by most subsequent writers. From a commercial standpoint this description is sufficient, for the land phosphate as mined generally occurs in small and more or less rounded fragments, and the river rock, which was extensively mined in the early days of the industry when the term was first used, is decidedly nodular in appearance. Geologically, however, the term is unfortunate, for it implies much in the way of genesis and history that is probably incorrect.

The two varieties of the phosphate, land rock and river rock, differ somewhat in character. The land rock occurs in a more or less irregular bed which represents the undisturbed phosphatized Edisto marl; the river rock probably consists in part of the original phosphatized Edisto and in part of fragments derived from the land deposits and

¹ Shepard, C. U., Jr., *South Carolina phosphates*, p. 20, Charleston, 1880.

concentrated on the river bottom, the whole deposit having been more or less worked over by river action. To these may be added a third kind, the Accabee gravels of Sloan, which are a later deposit formed by the reworking of the original bed.

The land rock occurs in a bed which has a maximum thickness of 30 inches, but averages probably between 8 and 16 inches, and is in many places absent. It generally consists of an irregular or even jagged mass of rock honeycombed by many tortuous cavities, but its upper surface is in many places smoother and more compact than the lower. In places the cavities are as much as 4 inches in diameter and thus afford lines of weakness which cause the bed to break up easily when disturbed in mining. The cavities are probably due chiefly to solution; their interiors are generally smooth, and when the rock is broken up a fragment resembles a very irregular nodule, the surface of which is partly smooth and partly rough and broken. Associated with the rock is a black sandy clay which contains a minor amount of small rounded phosphate nodules.

The term nodule was applied by the earlier writers to the whole mass, and even the large pieces removed in mining were described as "made up of nodules cemented together." Loosely scattered nodules are referred to by some of the older writers,¹ however, and it is probable that the solvent action was sometimes strong enough to enlarge the cavities until the mass was actually broken up. Some of the fragments are described as coated with a hard and lustrous enamel, but so far as can now be ascertained such fragments were of the river variety. No instance of concretionary structure in land rock is mentioned by any of the earlier writers. Moreover, the rock contains many casts and molds of the same species that characterize the unaltered Edisto marl,² and there seems little doubt that it is really a bed of irregularly phosphatized marl rather than a mass of concretionary nodules.

The rock varies considerably in color, hardness, and specific gravity, these characters being partly dependent on the degree of phosphatization. Much of the material now being mined is light yellowish brown in color and is soft and chalky. All variations in color from light brown to very dark gray are exhibited, and as the color deepens the hardness and specific gravity increase. A dark rock generally contains more lime phosphate per unit volume than a light-colored one and is therefore considered more valuable.

The cavities in the phosphate rock are filled with sandy clay, which is generally calcareous and somewhat phosphatic. Its color appears to depend largely on that of the rock with which it is associated. In

¹ Shaler, N. S., On the phosphate beds of South Carolina: Boston Soc. Nat. Hist. Proc., vol. 13, pp. 222-235, 1870.

² Sloan, Earle, op. cit., p. 299.

places this material contains quartz pebbles and pebbles or small nodules of phosphate rock. It also carries the abundant remains of many species of vertebrates which are not found within the rock mass itself.

The river rock occurs generally in loose and more or less rounded fragments of nodular aspect, which have accumulated in irregular banks on the river bottoms. In many localities they are well rounded and are inclosed by hard lustrous enamel. Penrose¹ distinguishes six different varieties of the river rock by their color and the presence or absence of enamel. The river rock is generally darker in color than the common land variety, being dark brown or gray or even jet-black. It is higher in specific gravity than the ordinary land rock and is generally harder, especially when inclosed by enamel. Some varieties are highly siliceous, the grains of sand being visible to the naked eye. According to Penrose some of the rock from Bull River exhibits a distinct concretionary structure, but this is very uncommon. In Morgan River and elsewhere the river rock has been found in a thin and more or less continuous bed like that of the ordinary land deposits, but it generally occurs in thick but irregular banks. Some of these deposits are fairly free from mud and sand and yield a product which requires little washing. In others the proportion of matrix is greater and may increase laterally until the phosphate is too sparsely scattered to allow mining. As would be expected in a river deposit, the distribution of the rock is generally very irregular, as is also the character of the overlying and underlying material. Where the river rock occurs as a more or less continuous and undisturbed bed it appears to rest upon the Cooper marl, but in its more common reworked condition it is generally separated from the marl by clay or sand.

Some writers have differentiated a third type which they call marsh deposits. These occur in a bed similar to the ordinary land deposits, but have the dark color and higher specific gravity of the river rock. They represent merely the undisturbed phosphate bed which has undergone certain changes. (See p. 209.)

The irregular deposits of phosphate and quartz pebbles known to the miners as the "Flying Rock bed" (the Accabee gravels of Sloan) are derived from the reworking of the original land deposits. They are chiefly of scientific interest, although at some places they have yielded a limited amount of phosphate.

¹ Penrose, R. A. F., jr., Nature and origin of deposits of phosphates of lime: U. S. Geol. Survey Bull. 46, pp. 60-70, 1888.

CHEMICAL COMPOSITION.

The commercial rock varies greatly in phosphate content, but at no place does it contain more than 64 per cent in the washed but unburned condition. Waggaman¹ states that isolated fragments have been found which contain as much as 75 per cent, but these do not occur in commercial quantity. Some of the rock mined in the Edisto district averages 62 per cent (see analysis 7, p. 196), but it is probable that the general average for the South Carolina area would approximate 58 per cent. There are many districts in which the phosphate content is as low as 55 or 53 per cent.

According to Shepard,² who made many hundred analyses of the South Carolina phosphate, the composition varies greatly in even one piece of rock. Thus the upper portion of the bed, which is generally flat and compact, is commonly somewhat higher in phosphate than the lower, more irregular, honeycombed portion. Similarly the exterior of a river nodule, especially when coated with enamel, may run 2 per cent or more higher in phosphate than the interior. The composition of the rock from a broad district is naturally even more variable, and irregular areas are sometimes found in mining in which the rock is of too low grade to work. The composition of the average commercial product, however, is fairly uniform.

In the following table are given fourteen analyses of the average commercial South Carolina product compiled from various sources, and three type analyses of the commercial Florida rock, inserted for comparison:

¹ Waggaman, W. H., Report on the phosphate fields of South Carolina: U. S. Dept. Agr. Bull. 18, p. 6, 1913.

² Shepard, C. U., jr., South Carolina phosphates, p. 21, Charleston, 1880.